

"Do not write anything on question-paper except Roll Number, otherwise it shall be deemed as an act of indulging in unfair means and action shall be taken as per rules."

Roll No. 22UMEC1155

BE II (ME)

3

Mat. Tech.

SECOND B.E. EXAMINATION - 2023

MECHANICAL ENGINEERING

(III SEMESTER CBCS)

ME - 202 A: Materials Technology (M)

Time - Three Hours

Maximum Marks - 80

Note:- (i) Attempt any **FIVE** questions.

(ii) Answer should be to the point.

(iii) All questions carry equal marks.

1. (a) Discuss imperfection in crystals with proper diagram. What are the advantages and disadvantage of imperfections. 8
- (b) Find out coordination number, atomic radius, no. of atoms in B.C.C., F.C.C. and H.C.P. structures. 6
- (c) Construct (101) and (110) plane with in unit cell. 6
2. (a) What are various types of dislocations in two dimension and three dimensions. 8
- (b) How dislocations are found out with the help of Berger Vector? 6
- (c) What are the cold and hot deformations. What factors responsible to affect them? 6

3. (a) Draw a neat iron carbon equilibrium diagram and label various regions of eutectic and eutectoid phase. 12
- (b) In continuation of C.I. equilibrium explain various structures so obtained with their mechanical properties. 8
4. (a) What do you understand by Heat treatment of steel? Why it is performed? What are the advantages and applications in various engineering skills.
- (b) Explain Annealing process. Discuss various types of Annealing methods.
5. (a) How hardness is imparted in steel. Explain various Quenching methods and Quenching media. 10
- (b) Discuss various Alloying elements in steel in brief. Write uses and applications of various types of carbon steel. 10
6. (a) Discuss various powder manufacturing methods. 10
- (b) How Sintering is performed? What are the advantages of sintering? Compare polymers, ceramics & composites. 10
7. Write short notes on (any four): 5X4
- (a) Mechanical properties of materials
 - (b) Fatigue and fracture
 - (c) Nitriding and cyaniding
 - (d) Brass, Bronze and Bearing material
 - (e) Smart materials and its applications

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Roll No.

B.E. II (ME)

4

Kine. of Mach.

SECOND B.E. EXAMINATION - 2023

MECHANICAL ENGINEERING

(III SEMESTER CBCS)

ME - 203 A: Kinematics of Machines - (M)

Time - Three Hours

Maximum Marks - 80

Note:- (i) Attempt any **FIVE** questions.

(ii) Q. No. 1 is compulsory & complete solution (including calculations) for Q. No. 1 is to be given in the drawing sheet.

(iii) All questions carry equal marks.

(iv) Assume suitable data, if not given and mention it clearly.

(v) Draw neat sketches, wherever necessary.

1. (a) Locate all the instantaneous centres of the slider crank mechanism as shown in Fig. 1. The length of crank $OB = 150$ mm and connecting rod $AB = 450$ mm, crank OB makes angle 45° with OA . Point C is located on connecting rod AB at 150 mm from point B . If the crank rotates clockwise with an

- angular velocity of 10 rad/second. Calculate the linear velocity of slider A, connecting rod AB and point C located on connecting rod AB. Also calculate the angular velocity and acceleration of the connecting rod AB. 16
2. (a) Write and draw all the possible the inversions of single-slider crank chain mechanism. 8
- (b) Distinguish between lower and higher pairs with examples and neat sketches. Explain various lower pairs with sketches. 8
3. (a) In a crank and slotted lever quick return motion mechanism, the distance between the fixed centres is 300 mm and length of driving crank is 150 mm. Find the inclination of the slotted bar with the vertical in the extreme position and time ratio of cutting stroke and to the return stroke. 8
- (b) Describe Grashof's law and Grubler's law. State how is Grashof's law helpful in classifying the four-bar mechanisms into different types. 8
4. (a) What is uniform pressure and uniform wear theories? Deduce expression for the friction torque considering both the theories for a flat collar. 8

- (b) A screw jack raises a load of 16 kN through a distance of 150 mm. The mean diameter and the pitch of the screw are 56 mm and 10 mm respectively. Determine the work done and the efficiency of the screw jack when the:
- Case 1. Load rotates with the screw.
- Case 2. Loose head on which the load rests does not rotate with the screw and the outside and the inside diameters of the bearing surface of the loose head are 50 mm and 10 mm respectively. Take coefficient of friction for the screw and the bearing surface as 0.11. 8
5. (a) Define and elaborate the law of belting. Derive the relation for ratio of belt tensions in a flat-belt drive. 8
- (b) In a belt drive, the mass of the belt is 1.5 kg/m length and its speed is 6 m/second. The drive transmits 12 kW of power. Determine the initial tension in the belt and the strength of the belt. The coefficient of friction is 0.25 and the angle of lap is 165° . 8
6. (a) A rope drive is required to transmit 250 kW from a sheave of 1 m diameter running at 450 rpm. The safe pull in each rope is 800 N and the mass of the

rope is 0.46 kg/m length. The angle of lap is 160 degree and the groove angle is 45 degree. If the coefficient of friction is between the rope and the sheave is 0.3; calculate the number of rope. 8

- (b) Explain the terms slip and creep as referred to belt drive. On what factors does it depend? Explain how creep affects the power transmission capacity of the belt drive. 8

7. (a) Explain watt's mechanism with neat sketch. Show, how can draw an approximate straight line? 8

- (b) A Hook's joint connects two shafts whose axes intersect at 20° . The driving shaft rotates at a uniform speed of 300 rpm. The driven shaft with attached masses has a mass of 60 kg and radius of gyration of 120 mm. Determine the:

- (a) Torque required at the driving shafts if a steady torque of 180 Nm resists rotation of the driven shaft and the angle of rotation is 45° .

- (b) Angle between the shafts at which the total fluctuation of speed of the driven shaft is limited to 25 rpm. 8

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Roll No. 22VME11155

B.E. II (ME)

3

Maths.

SECOND B.E. EXAMINATION - 2023

MECHANICAL ENGINEERING

(III SEMESTER CBCS)

ME - 205 A: Mathematics (M)

Time - Three Hours

Maximum Marks - 80

- Note:- (i) Attempt any **FIVE** questions.
(ii) Answer should be to the point.
(iii) All questions carry equal marks.

1. (a) Solve $\frac{dx}{dt} - 7x + y = 0$; $\frac{dy}{dt} - 2x - 5y = 0$ 8
(b) Solve $(2x^2 + 2xy + 2xz^2 + 1) dx + dy + 2zdz = 0$ 8
2. (a) Solve the following partial diff. equation by using Charpit's Method
 $2xz - px^2 - 2qxy + pq = 0$ 8
(b) Solve the following partial differential equation by using Monge's method
 $x^2r - 2xs + t + q = 0$ 8

3. (a) The area A of a circle of diameter d is given for the following values. 8

d :	80	85	90	95	100
A:	5026	5674	6362	7088	7854

Find approximate values for the areas of circle of diameter 82.

- (b) Use Stirling formula to find Y_{28} , given $Y_{20} = 49225$, $Y_{25} = 48316$, $Y_{30} = 47236$, $Y_{35} = 45926$, $Y_{40} = 44306$ 8

4. (a) Use Simpson's 1/3 and 3/8 rule to evaluate the following: 8

$$\int_0^1 \frac{dx}{1+x^2}$$

Hence obtain the approximate value of π in each case.

- (b) Find the root of the equation $2x - \log_{10} x = 7$ which lies between 3.5 and 4 by Regula falsi method. 8

5. (a) Prove that $L\left\{\frac{\sin t}{t}\right\} = \tan^{-1}\left(\frac{1}{s}\right)$

and hence find $L\left\{\frac{\sin at}{t}\right\}$ 8

- (b) Solve by using Laplace transform 8

$$(D^2 + 9)y = \cos 2t$$

given that $y(0) = 1$, $y(\pi/2) = -1$

6. (a) Find the Fourier Sine and Cosine transform of 8

$$f(x) = \begin{cases} 1 & ; 0 < x < a \\ 0 & ; x > a \end{cases}$$

2

(Contd.)

- (b) Find the Fourier Sine transform of e^{-x} ; $x > 0$ 8

Hence show that $\int_0^\infty \frac{x \sin mx}{1+x^2} dx = \frac{\pi}{2} e^{-m}$

7. (a) Find the value of $f(5)$ from the following table by using Lagrange's interpolation formula 8

x:	1	2	3	4	7
f(x):	2	4	8	16	128

- (b) Using Runge - Kutta method of fourth order, find an approximate value of $y(0.2)$ and $y(0.4)$ by solving

$$\frac{dy}{dx} = -2xy^2 \text{ with } x = 0, y = 1; h \text{ being } 0.2 \quad 8$$

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Roll No. 22U.MEC.1155

BE II (ME)

3

Engg. Thermo.

SECOND B.E. EXAMINATION, 2023
(MECHANICAL ENGINEERING)
(III SEMESTER CBCS)

ME – 201 A: ENGINEERING THERMODYNAMICS (M)

Time – Three Hours

Maximum Marks – 80

- Note: (i) Attempt any **Five** questions.
(ii) Answer should be to the point.
(iii) All questions carry equal marks.

1. (a) In a reciprocating engine, the mass of gas occupying the clearance volume is m_c kg at state p_1, u_1, v_1 , and h_1 . By opening the inlet valve, m_f kg of gas is taken into the cylinder, and at the conclusion of the intake process the state of the gas is given by p_2, u_2, v_2 , and h_2 . The state of the gas in the supply pipe is constant and is given by p_p, u_p, v_p, h_p and V_p . How much heat is transferred between the gas and the cylinder walls during the intake process? 8
- (b) State the Kelvin- Planck statement of second law of thermodynamics. Write down the corollaries of second law of thermodynamics. 8
2. (a) Derive the expression of exergy for a steady flow system (open system). 8
- (b) 4 kg of water at 50°C are mixed with 5 kg of water at 100°C in a steady flow process. Calculate:
 - (i) the temperature of resulting mixture
 - (ii) the change in entropy
 - (iii) the unavailable energy with respect to the energy receiving water at 50°C . 8

3. (a) Derive the vander Waals constants in terms of critical properties.

$$a = 3p_c \cdot v_c^2, b = \frac{1}{3}v_c \text{ \& } R = \frac{8a}{9T_c v_c} \quad 8$$

- (b) State Dalton's law of partial pressure, Amagat's law of additive volumes and gibbs theorem. 8

- 4 (a) Write down the first and second Tds equations. Derive the expression for the difference in heat capacities, C_p & C_v . What does the expression signify? 8

- (b) Explain joule kelvin effect using T-p diagram. Mention inversion curve, cooling and heating region on the T-p diagram. Derive the value of joule kelvin coefficient for ideal gas. 8

5. (a) Derive the expression of work done per cycle for single stage reciprocating air compressor. 8

- (b) A multi stage air compressor takes in air at 1 bar, 298 K and delivers at 36 bar. The maximum temperature in any stage is not to exceed 390 K. If the law of compression and expansion is $pv^{1.3} = \text{constant}$, find the number of stages for minimum power input. Estimate the power required. 8

6. (a) What is stoichiometric air fuel ratio and excess air? A fuel having formula C_8H_{18} is burnt with chemically correct amount of air at atmospheric pressure. Write the reaction equation and find the stoichiometric air fuel ratio. 8

- (b) Describe the orsat apparatus with a sketch. Explain how the gases are analysed with the help of this apparatus. 8

7. Write short notes on any four- 4x4=16

- (i) International practical temperature scale.
- (ii) Classification of air compressors
- (iii) Second law efficiency
- (iv) Carnot cycle
- (v) Effect of pressure ratio and clearance on volumetric efficiency of reciprocating compressor
- (vi) Maxwell's equations.

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Roll No. 22MECE1155...

BE II (ME)

3

Mech. Meas. & Inst.

SECOND B.E. EXAMINATION, 2023

(MECHANICAL ENGINEERING)

(III SEMESTER CBCS)

**ME-206 A: MECHANICAL MEASUREMENTS AND
INSTRUMENTATION (M)**

Time – Three Hours

Maximum Marks – 80

- Note: (i) Attempt any **Five** questions.
(ii) Answer should be to the point.
(iii) All questions carry equal marks.

1. (a) What is a transducer, discuss its various types and desired characteristic. 6
(b) Explain Null and deflection methods of measurement. 4
(c) Draw the block diagram of a generalised measurement system. Identify the various elements and point out the function performed by each element. 6
2. (a) Explain the term impedance loading and matching of a generalised measurement system. Derive the expression for transmitting maximum power from the transducer to the external load. 8
(b) The following readings are taken of a certain are physical length with the help of a micrometer Screw.
1.41 1.45 1.53 1.54 1.49 1.51 1.60 1.55 1.47 and
1.66 mm
Assuming that only random errors are present. Calculate the arithmetic mean, the average deviation, standard deviation, variance and the probable error of the reading. 8

3. (a) Sketch and explain a thermocouple vacuum gauge with its merits and demerits. 8
- (b) A Wheatstone bridge circuit comprises four resistance; One resistance is connected in each limb the bridge. Under null mode operation, the solution of the unknown resistance is

$$R_u = \frac{R_2 \times R_3}{R_1}$$

The known resistance and the limiting errors associated with them are:

$$R_1 = (100 \pm 0.75\%) \text{ ohm}$$

$$R_2 = (1000 \pm 0.5\%) \text{ ohm}$$

$$R_3 = (850 \pm 0.6\%) \text{ ohm}$$

Determine the magnitude of unknowns resistance K_u and its limiting errors. Also workout the guaranteed value of the resistance. 8

4. (a) Sketch a typical optical pyrometer. Explain its working and list its notable characteristic. 8
- (b) The hot junction of an iron- constantan thermocouple is connected to a potentiometer whose terminals are at 25°C. The potentiometer reads 3.6 mV. Work-out the temperature at the hot junction of the thermocouple. It may be assumed that the couple conforms to the following values which are based reference junction at 0 °C.

Temperature	-20	10	40	70	100	130
Voltage mV	-0.8	0.5	1.5	3.41	4.91	6.42

5. (a) List Various flow measuring devices. Sketch and explain the working of Magnetic flow meter. 8
- (b) State the objective of flow-visualisation. Explain some of the methods commonly adopted for flow visualisation in low speed flow. 8

- 6 (a) Describe the principle of operation variable of linear variable differential transformer (LVDT). Sketch the typical input and output graph. 8
- (b) Explain the function of a dummy gauge in a strain gauge load cell. Draw the circuit arrangement of a bridge employing two strain gauges for the measurement of strain of a cantilever beam so that no error is caused because of change in gauge resistance due to environmental temperature change. 8

- 7 Write short notes on any four -

- Use of strain gauge on rotating shaft
- CRO
- Bridgeman Gauge
- RTD & thermister
- pneumatic load cell
- Reed Vibrometer

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B.E. II (ME)

3

Mech. of Soli.

SECOND B.E. EXAMINATION - 2023

MECHANICAL ENGINEERING

(III SEMESTER CBCS)

ME - 204 A: Mechanics of Solids (M)

Time - Three Hours

Maximum Marks - 80

Note:- (i) Attempt any Five questions.

(ii) Answer should be to the point.

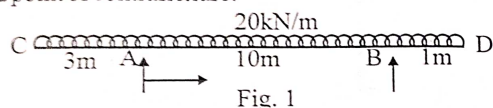
(iii) All questions carry equal marks.

1. (a) A load of 2 MN is applied on a short concrete column $500 \text{ mm} \times 500 \text{ mm}$. The column is having 4 steel bars of 10 mm diameter one at each corner. Find the stresses in the concrete and steel bars. Young's modulus for steel is $2 \times 10^5 \text{ N/mm}^2$ and for Concrete is $1.4 \times 10^4 \text{ N/mm}^2$.

8

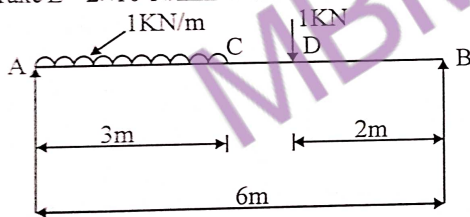
- (b) If the tension test bar is found to taper from $(D+a)$ to $(D-a)$ diameter, prove the error involved in using mean diameter to calculate Young's modulus is $(10a/D)$ percent. 8

2. Draw SFD and BMD for the beam shown in Fig. 1. Also find point of contraflexure. 16



3. (a) Explain theory of simple bending and derive bending equation stating the assumptions. 12
 (b) Why the I-section is better than other sections for a beam? 4
4. A simply supported beam is loaded as shown in Fig 2. Determine the deflection at points C and D in the beam. 16

Take $E = 2 \times 10^5 \text{ N/mm}^2$ and $I = 2 \times 10^8 \text{ mm}^4$.



5. (a) Derive torsion equation stating the assumptions. 8
 (b) A bar of steel has rectangular cross section $30 \times 20 \text{ mm}$. Find the percentage decrease of area of cross section, when it is subjected to a tensile force of 120 kN in the direction of its length. Take $E = 2 \times 10^5 \text{ N/mm}^2$ and $\mu = 0.3$. 8
6. (a) A steel bar of rectangular section $30 \text{ mm} \times 40 \text{ mm}$ one end fixed and other is free at each end is subjected to axial compression. The bar is 1.75 m long. Determine the buckling load and the corresponding axial stress using Euler's formula. Determine the minimum length for which Euler's equation may be used to determine the buckling load, if the proportional limit of the material is 200 N/mm^2 . Take $E = 2 \times 10^5 \text{ N/mm}^2$ 16
7. (a) At a point in a bracket the stresses on two mutually perpendicular planes are 240 N/mm^2 tensile and 120 N/mm^2 tensile. The shear stress across these planes is 60 N/mm^2 . Find the principal stresses and maximum shear stress at the point. 12
 (b) Explain limitation of Euler's theory of column. 4